C. KARTHIK RAJA

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CAREER OBJECTIVE

AI/ML Engineer who turns AI requirements into working systems with clear, usable results. I bring hands-on experience building speech-to-text emotion detection pipelines, RAG-powered research tools, and medical imaging classifiers, along with leadership as Student Council President — so organizations get deployable AI solutions and someone who can align technical work with team and organizational goals.

EDUCATION

B.E. Computer Science and Engineering Expected 2027
Nehru Institute of Technology (Autonomous) CGPA 8.67 · through 6th semester

Higher Secondary — Class XII (CBSE) 2022
Sri Ramakrishna Public School 78.6%

Secondary — Class X (CBSE) 2020
Sri Ramakrishna Public School 89.9%

TECHNICAL SKILLS

Languages: Python 3, C
Web Development: HTML, CSS, Bootstrap, Django
AI Foundations: Machine Learning, Deep Learning, Neural Networks, Genai(Learning)
Libraries & Frameworks: NumPy, Pandas, DistilBERT
Databases & APIs: MySQL
Tools: Git, GitHub, Docker, VS Code, PyCharm, Jupyter Notebook
Soft Skills: Communication, Quick Learning, Logical Thinking

PROJECTS

News Research Tool *Python · LangChain · RAG · FAISS · Ollama · Streamlit*

- Built a RAG-based news research application that processes article URLs and enables context-aware question answering.
- Implemented document chunking, HuggingFace embeddings, FAISS similarity search, and retrieval-based context generation.
- Integrated Ollama Llama 3.2 for local LLM inference and developed a Streamlit interface with source references.

AI Mental Health Assistant *Python · TensorFlow · DistilBERT · FastAPI · React.js · MySQL · Whisper*

- Fine-tuned DistilBERT for emotion classification from text and built an end-to-end AI inference pipeline covering preprocessing, prediction, confidence estimation, and personalized CBT-inspired response generation.
- Developed a FastAPI backend with JWT authentication and MySQL integration for secure user management and persistent chat history; integrated OpenAI Whisper for voice and text interaction.
- Built a React frontend with dark/light themes, emotion analytics dashboard, wellness points, streak tracking, and progress visualization to drive user engagement.

ATM Transaction System *Python · MySQL · Twilio API*

- Developed a console-based ATM system supporting deposit, withdrawal, and balance inquiries.
- Integrated MySQL for secure transaction storage and Twilio API for real-time SMS notifications.
- Implemented user authentication and session management to ensure transaction security.

PRESENTATIONS & CONTESTS

- **1st Place** — Pen & Program coding event, TechGeeks 2025, Dhaanish Ahmed Institute of Technology, Coimbatore (8 April 2025).
- **3rd Place** — "AI Powered Rail Safety for Public Benefit," TechGeeks 2025 National Level Symposium, Dhaanish Ahmed Institute of Technology, Coimbatore (8 April 2025).
- **Presented** "AI Powered Rail Safety for Public Benefit" at Tech Magnetz 2024 National Level Technical Symposium, Hindusthan Institute of Technology, Coimbatore (26 March 2024).

CERTIFICATIONS & WORKSHOP

- **Machine Learning (Supervised)** by Stanford University, May 2026.
- **Python for Data Science, AI & Development** by IBM.
- **Oracle Certified AI Foundation Associate** — Oracle University.

EXTRA-CURRICULAR ACTIVITIES

- **Student Council President** — Nehru Institute of Technology (2026–2027). Elected to lead the Student Council, coordinate student activities, and represent student interests while collaborating with college administration.
- Active participant in national-level technical symposia and inter-college coding competitions.
- Engaged in collaborative academic projects, demonstrating teamwork and communication skills.
- Continuously upskilling through Coursera, NPTEL, and Oracle University.

PERSONAL DETAILS

Date of Birth: 04 April 2006
Father's Name: A. Chinnadurai
Languages Known: Tamil, English
Address: 8/174, West Street, T. Kudikadu, Thathanur Post, Udayarpalayam, Ariyalur

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief. I take full responsibility for the authenticity of all details mentioned in this resume.

Place:

Date:

C. Karthik Raja